Hyperspectral Image Super-Resolution Based on Adapter-Enhanced Foundation Models

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***Abstract*—Hyperspectral images (HSIs) contain rich spectral information and are highly valuable in applications such as remote sensing monitoring and target identification. However, due to limitations in imaging technology and cost constraints, existing HSIs generally suffer from low spatial resolution. Recently, several hyperspectral vision foundation models have been proposed, employing large-scale network architectures and newly constructed large-scale datasets, achieving remarkable performance in HSI super-resolution (SR) and other hyperspectral vision tasks. However, as these general-purpose foundation models are designed to accommodate diverse vision tasks, their architectures exhibit inherent limitations when applied specifically to SR as a low-level vision task. To address this architectural limitation, we propose HyperAdapter, an enhanced network based on HyperSIGMA that employs task-specific architectural optimization to fully exploit the performance potential of foundation models for SR tasks. Specifically, we introduce a spatial–spectral prior module (SSPM) to improve the model's ability to capture local spatial context and spectral correlations in HSI, along with an adapter module to effectively integrate and refine the spatial and spectral features of the base model, ensuring better adaptation to the HSI SR task. Experiments on three public HSI SR datasets demonstrate that the proposed method significantly improves reconstruction performance, validating its superiority and practical utility.**

***Index Terms*—Adapter-enhanced foundation models, hyperspectral image (HIS), super-resolution (SR).**

I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral information at each pixel, making them widely used in various fields such as environmental monitoring [1], [2], resource exploration [3], [4], and object recognition [5], [6] [7]. Despite offering more detailed material information compared to traditional RGB images, HSIs often suffer from low spatial resolution due to limitations in current imaging technologies, which significantly restricts their effectiveness in practical applications. Therefore, improving the spatial resolution of HSIs has become a key research challenge in remote sensing image processing.

Traditional HSI super-resolution (SR) methods, such as interpolation, sparse representation [8], [9], [10], and low-rank matrix decomposition [11], [12], [13], rely heavily on prior knowledge of the image. Although they can improve resolution to some extent, they often fail to recover fine details in complex scenes and do not fully leverage the spatial–spectral joint characteristics of the image. In recent years, with the rapid development of deep learning technologies, deep learning-based HSI SR methods have gradually become a research hotspot. These methods leverage the powerful feature extraction capabilities and end-to-end optimization strategies of deep learning to effectively learn high-level spatial–spectral features from data, significantly improving the SR reconstruction performance of HSI. For example, Ran et al. [14] innovatively proposed kernel space nonlocal convolution (KNLConv), which enhances HSI SR performance by exploring global dependencies in kernel space and introducing adaptive pointwise convolution. Li et al. [15] innovatively proposed a multistage model-informed unsupervised network (M2U-Net), which significantly enhances HIS SR performance through a three-stage unsupervised framework that integrates degradation models and deep image priors, combined with a dual U-Net and cross-attention mechanism. Li et al. [16] proposed an enhanced deep image prior network (EDIP-Net) that achieves high-quality unsupervised HSI SR through a two-stage learning scheme and a double U-shaped architecture. These methods have demonstrated powerful performance in HSI SR tasks, driving the advancement of this field.

Recently, several hyperspectral vision foundation models [17], [18] have demonstrated remarkable performance across various hyperspectral vision tasks, including HSI SR, employing large-scale network architectures and newly constructed large-scale datasets. However, as these general-purpose foundation models are designed to accommodate diverse vision tasks, their architectures exhibit inherent limitations when applied specifically to SR, a low-level vision task. Taking HyperSIGMA [18] as an example, although its dual-branch parallel architecture (spatial feature extraction branch and spectral feature learning branch) performs well in general tasks, this design struggles to fully facilitate deep interaction

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between spatial and spectral features in SR tasks, thereby constraining the model's performance potential.

To address this architectural bottleneck, we propose HyperAdapter, an enhanced network based on HyperSIGMA [18] that achieves full exploitation of the base model's performance potential through task-specific optimization tailored for SR. Specifically: 1) a spatial–spectral prior module (SSPM) is designed to effectively improve SR reconstruction performance by enhancing the model's ability to capture local spatial context and spectral correlations and 2) an adapter module is developed to deeply integrate and optimize the spatial and spectral features of the base model, ensuring a better adaptation to the specific requirements of HSI SR tasks.

This improvement preserves the powerful feature representation capabilities acquired by the base model through large-scale dataset training while overcoming the limitations of general-purpose models in SR tasks via task-oriented architectural optimization. In summary, our contributions are as follows.

1) We propose the HyperAdapter enhancement network based on HyperSIGMA, which addresses the insufficient feature interaction caused by architectural limitations of general hyperspectral foundation models in SR tasks through task-specific architectural optimization, significantly improving reconstruction performance.
2) We propose an efficient adapter module that deeply integrates the spatial and spectral branch features of HyperSIGMA through spatial–spectral coupled attention module (SSCAM), token fusioner (CTF), and token injector (CTI) while optimizing local context awareness via spatial–spectral prior modeling, thereby significantly improving the model's reconstruction performance.
3) Extensive experimental validation on several publicly available HSI SR datasets, demonstrating that the proposed method significantly outperforms existing approaches.

II. Related Work

A. HSI SR

With the advancement of deep learning, deep learning-based methods have become the predominant approach for HSI SR. Unlike traditional optimization-based methods, deep learning approaches construct specialized network architectures to learn nonlinear mappings between low-resolution (LR) and high-resolution (HR) images using large-scale training samples in an end-to-end manner.

Among existing architectures, 2-D CNNs were first applied to HSI SR. Li and Han [19] pioneered the application of deep 2-D CNNs in HSI reconstruction. To better exploit spectral information, Jiang et al. [20] incorporating spectral attention to capture interband correlations with progressive upsampling. Liu et al. [21] combine spectral grouping with attention mechanisms and Wang et al. [22] employ recursive feedback to capture cross-band characteristics with spatial–spectral regularization. Considering the intrinsic 3-D structure of HSI, 3-D CNNs have also been introduced for SR tasks. Mei et al. [23] used 3-D CNNs to reduce spectral distortion from 2-D CNNs. However, 3-D CNNs typically exhibit higher computational complexity and memory requirements. To address this, Li et al. [24] investigated 2-D/3-D convolution relationships to reduce complexity. Liu et al. [25] proposed a 2-D/3-D dual-domain network that synergistically extracts spatial–spectral and frequency features via progressive upsampling. Hu et al. [26] reformulated HSI SR in the abundance domain using spatial-preserving decomposition to estimate abundance representations with spatial correlations. Chen et al. [27] fused multidomain features via spatial–spectral attention. Wang et al. [28] proposed MESR with parallel spectral maintenance and context affinity enhancement to boost performance while reducing memory. Qi et al. [29] proposed a semi-supervised framework with degradation-aware pretraining to achieve HSI SR.

Recent advances in HSI SR have been driven by network architecture optimization. Liu et al. [30] constructed a coupled network integrating 2-D CNN and Transformer branches to extract complementary spatial–spectral features. Zhang et al. [31] developed a Transformer-based iterative refinement architecture for multiscale context learning. Yu et al. [32] adopted a dual-stream structure with spatial and spectral attention mechanisms. Chen et al. [33] introduced multiscale feature extraction with deformable attention. In addition, Xu et al. [34] proposed MambaHSISR, which achieves HSI SR through a novel Mamba-based framework integrating spatial and spectral state-space modeling. Pang et al. [35] proposed HIR-Diff, which achieves unsupervised HSI restoration through an improved diffusion model. Beyond network architecture optimization, band grouping strategies have also become a widely adopted technical approach among researchers. Wang et al. [36] implemented adjacent/nonadjacent band grouping processing via a bidirectional network. Zhang et al. [37] employed a Dual ODE method to achieve adaptive band correlation based on an ODE dynamical system. Cheng et al. [38] realized cross-domain adaptive reconstruction through a progressive band expansion strategy within a meta-transfer learning framework.

Despite these advancements, existing methods still face two critical bottlenecks: 1) due to insufficient network capacity, previous SR models struggle to fully characterize the complex spatial–spectral features of HSIs and 2) the lack of large-scale open-source datasets hinders the adequate training of deep networks, limiting the models' generalization capabilities.

Recently, some hyperspectral vision foundation models [17], [18] have achieved outstanding results in diverse hyperspectral vision tasks, including HSI SR, through the use of large-scale architectures and newly developed datasets. However, as these general-purpose foundation models are designed to accommodate diverse vision tasks, their architectures inherently lack specialization for SR tasks, which belong to low-level vision processing.

B. Remote Sensing Foundation Models

Remote sensing foundation models are large-scale pretrained models that provide robust starting points for downstream tasks in the remote sensing domain. By leveraging

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massive datasets and advanced architectures, these models can effectively capture complex surface features and spatiotemporal patterns, requiring minimal additional training to quickly adapt to specific application scenarios.

Based on differences in pretraining strategies, existing remote sensing foundation models can be primarily categorized into supervised and unsupervised types. Supervised pretraining methods rely on large-scale annotated datasets for model training, such as Wang et al. [39], who utilized the Million-AID dataset [40] to pretrain CNN and ViT models. However, the annotation process for remote sensing data is not only time-consuming and labor intensive but also costly. Meanwhile, with the rapid development of remote sensing technology, massive amounts of unannotated remote sensing data are continuously accumulating. This situation has prompted researchers to turn their attention to unsupervised pretraining methods to fully exploit the potential value of unannotated data. Among unsupervised pretraining methods, contrastive learning and masked image modeling are two mainstream technical approaches. Contrastive learning methods [41], [42] leverage the inherent characteristics of remote sensing data, such as seasonal variations [43], temporal evolution [44], and geospatial correlations [45], by constructing positive and negative sample pairs for training. However, these methods often require carefully designed pretext tasks and complex data augmentation strategies, which not only increase the difficulty of model design but also raise training costs, limiting their widespread application in practice.

To overcome the limitations of contrastive learning methods, masked image modeling methods [46], [47], [48] have garnered significant attention in recent years. These methods learn feature representations by predicting masked regions of images, which exhibit stronger self-supervised properties. In the field of remote sensing, SatMAE [49] was the first to extend the MAE framework to multispectral data processing, while DOFA [50] further enhanced model performance by introducing neural plasticity feature extraction methods.

Recently, several remote sensing foundation models [17], [18] have achieved breakthrough progress in various remote sensing vision tasks by constructing ultralarge-scale training datasets and stacking deep network parameters. However, these models typically adopt a unified architecture design to accommodate multitask requirements, which may not fully unleash their potential in specific tasks such as SR. To address this limitation, this article proposes an adapter-enhanced improvement scheme that breaks through the original architectural constraints via modular extensions, significantly enhancing the performance of foundation models in SR tasks.

III. Methodology

A. Overall Framework of the Proposed HyperAdapter

Recently, with the advancement of deep learning techniques, several hyperspectral vision foundation models have been proposed and demonstrated remarkable performance across various HSI processing tasks. However, as these general-purpose foundation models are designed to accommodate diverse vision tasks, their network architectures exhibit inherent limitations when specifically applied to SR as a low-level vision task. Taking the state-of-the-art HyperSIGMA [18] foundation model as an example, our in-depth architectural analysis reveals that its spatial and spectral feature branches adopt a completely parallel and mutually independent processing paradigm, lacking effective cross-modal feature interaction mechanisms. This architectural deficiency hinders the model from achieving optimal spatial–spectral feature co-representation, thereby significantly constraining its performance in HSI SR tasks.

To address this gap, we propose a HyperAdapter network specifically designed to enhance spatial–spectral feature fusion. Through carefully engineered modules, our network promotes synergistic integration of spatial and spectral information, strengthening cross-branch collaboration. The HyperAdapter network introduces two new core components to the HyperSIGMA [18] framework: the SSPM and the adapter module.

The overall architecture is illustrated in Fig. 1, with the modules we proposed highlighted in orange boxes. The processing pipeline begins with the input image being simultaneously fed into both original feature branches of HyperSIGMA [18] and the SSPM. After extracting corresponding local feature information, the obtained features are then input into the adapter module for spatial–spectral feature fusion. In this module, the number of adapters (N) serves as a hyperparameter—through comprehensive evaluation of efficiency and effectiveness, we set the interaction count $N = 4$. Accordingly, the encoder in HyperSIGMA [18] is evenly divided into N blocks, with each block containing $1/N$ encoder layers. These encoder blocks are interconnected via adapters.

For the ith adapter, the input consists of three components: the output from the previous adapter, along with the outputs from the ith spatial feature branch encoder block and the ith spectral feature branch encoder block in HyperSIGMA [18]. After processing through the adapter, three corresponding outputs are generated: one serves as input to the $(i + 1)$th adapter, while the other two feed into the $(i + 1)$th spatial and spectral feature branches.

The design of adapter enables effective integration of spatial and spectral feature information, significantly enhancing feature interaction performance. Finally, the outputs from both spatial and spectral feature extraction branches are jointly processed by the spectral enhancement module to further refine spectral characteristics, ultimately enabling high-quality image reconstruction through the upsampling module. In Sections III-B and III-C, we will provide a detailed description of the SSPM and the adapter module of our method.

B. Spatial–Spectral Prior Module

The SSPM serves as the core component of the HSI SR method proposed in this article. Its design aims to overcome the limitations of traditional approaches that process spatial and spectral features separately. By constructing a spatial–spectral coupling mechanism and adopting a hybrid architecture that combines convolutional operations with self-attention mechanisms, the module effectively models the complex relationships between spatial and spectral features.

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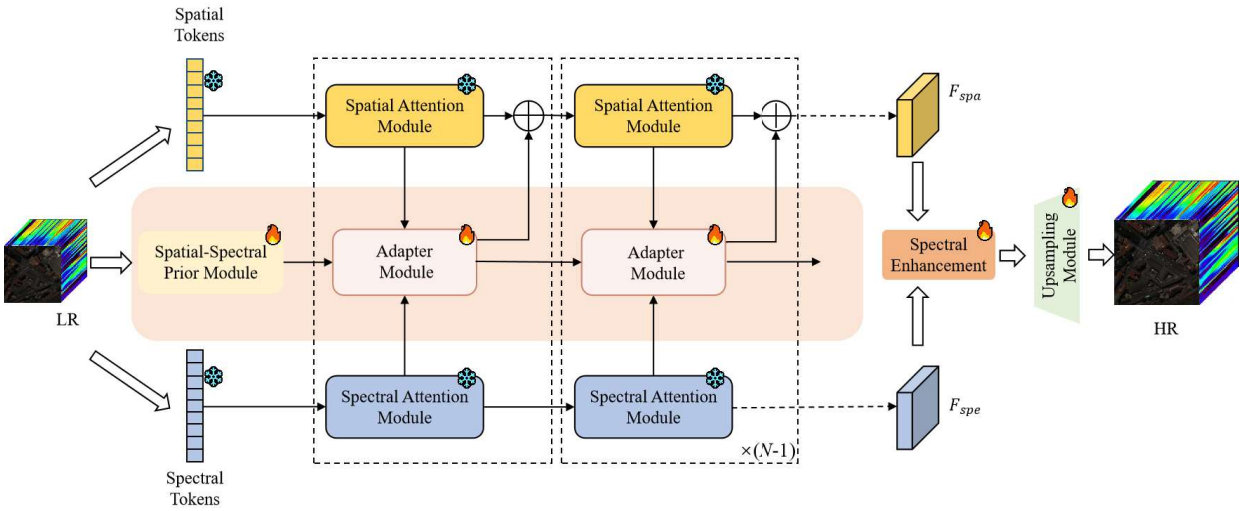


Fig. 1. Overall architecture of our proposed HyperAdapter network is built upon the dual-branch structure of the baseline HyperSIGMA model. Specifically, we design an enhancement component (highlighted by the orange box in the figure) that is placed between the upper spatial feature branch and the lower spectral feature branch. This component consists of a spatial–spectral prior module followed by N cascaded adapter modules, where N denotes the total number of adapter modules and the term $\times(N-1)$ indicates that these adapter modules are repeated $(N-1)$ times. The optimized spatial–spectral features are then fed into a feature-fusion module and an upsampling module to produce the final SR result.

Specifically, it employs multiscale convolutional kernels to extract local spatial features in parallel, capturing detailed image information efficiently. Meanwhile, a spectral attention mechanism is utilized to model long-range dependencies among spectral bands. Finally, feature reorganization and fusion strategies are applied to establish deep interconnections between spatial and spectral dimensions. This design enables the module to collaboratively model the complex nonlinear relationships between spatial structures and spectral characteristics, thereby significantly improving both the reconstruction quality and spectral fidelity of the images.

In the proposed module, the input LR HSI I_{LR} is first divided into G groups, each group containing c spectral bands and with an overlap of o bands between adjacent groups. Specifically, the input LR HSI is represented as

$$\left[I_{\mathrm{LR}}^{1}, I_{\mathrm{LR}}^{2}, \ldots, I_{\mathrm{LR}}^{g}, \ldots, I_{\mathrm{LR}}^{G}\right] = \mathrm{Divide}\left(I_{\mathrm{LR}}\right) \tag{1}$$

where I_{LR}^{g} denotes the gth group of images, each containing c spectral bands, and the groups have overlapping bands.

The core of the SSPM is the multiscale spatial–spectral attention module (MSSAM), which extracts multiscale spatial features using convolution kernels of varying sizes and captures spectral dependencies using a spectral attention mechanism. For the gth group of images, the spatial feature extraction process can be represented as

$$\begin{aligned} F_{\mathrm{local}}^{g} &= \mathrm{MSSAM}\left(I_{\mathrm{LR}}^{g}\right) \\ &= W_{1\times1}\left(I_{\mathrm{LR}}^{g}\right) + W_{3\times3}\left(I_{\mathrm{LR}}^{g}\right) + W_{5\times5}\left(I_{\mathrm{LR}}^{g}\right) \end{aligned} \tag{2}$$

where MSSAM$(\cdot)$ denotes the multiscale spatial–spectral attention module operation and $W_{d\times d}$ are the convolutional weights for the different kernels. The features of all groups are concatenated along the spectral dimension to obtain the overall feature

$$F_{\mathrm{local}} = \mathrm{Cat}\left(F_{\mathrm{local}}^{1}, F_{\mathrm{local}}^{2}, \ldots, F_{\mathrm{local}}^{g}, \ldots, F_{\mathrm{local}}^{G}\right) \tag{3}$$

where Cat$(\cdot)$ represents concatenation along the spectral dimension. The concatenated features are then passed to the spectral enhancement module for further processing. To integrate global information, the local features F_{local} are mapped to the hyperspectral dimension through a 1×1 convolution operation

$$F_0 = H_0\left(F_{\mathrm{local}}\right) \tag{4}$$

where $H_0(\cdot)$ represents the 1×1 convolution operation, and the output $F_0 \in \mathbb{R}^{H\times W\times C}$ represents the fused spatial–spectral features, with H and W being the spatial dimensions of the image and C being the spectral dimension.

Through this design, the SSPM enables the model to effectively enhance both spatial and spectral feature interactions, utilizing multiscale convolutions and spectral attention mechanisms. Not only does this module improve the recovery of image details, but it also strengthens the spatial–spectral dependencies in HSIs. The module provides high-quality spatial and spectral feature inputs for the subsequent adapter module, significantly improving the SR reconstruction quality.

C. Adapter Module

The adapter module is designed to effectively integrate and optimize features from the spatial and spectral branches of the base model, HyperSIGMA [18], addressing the inefficiency in feature collaboration between different branches when processing HSI SR tasks. This enables the model to better focus on the intricate characteristics of HSIs. Each adapter consists of three main modules, as shown in Fig. 2: the SSCAM, the token fusioner (CTF), and the token injector (CTI). In the jth adapter, in addition to the output feature F_{ada}^{j} from the previous adapter, the output features F_{spa}^{j} from the jth spatial sparse sampling attention module in the spatial feature extraction branch of HyperSIGMA [18] and F_{spe}^{j} from the jth spectral sparse sampling attention module in the spectral feature extraction branch are also used as inputs.

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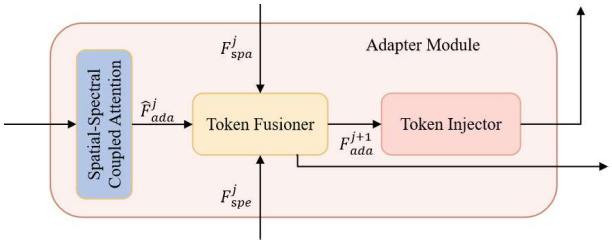


Fig. 2. Architectures of the adapter module.

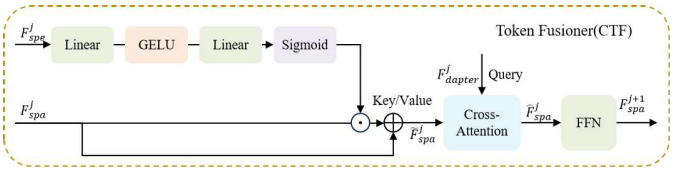


Fig. 3. Architectures of the token fusioner module.

First, F^j_{ada} is passed into the SSCAM to model the long-range dependencies between the spatial and spectral dimensions. The output $\hat{F}^j_{\text{ada}}$ is processed by the spatial–spectral feature extraction, as shown in the following equation:

$$\hat{F}^j_{\text{ada}} = \text{SSCAM}_j\left(F^j_{\text{ada}}\right). \tag{5}$$

After the spatial–spectral feature extraction, the next step is the token fusioner module CTF_j, which effectively fuses the extracted spatial–spectral features from HyperSIGMA [18] into the adapter, thus facilitating the integration and optimization of multimodal features. The output feature F^{j+1}_{ada} is given by

$$F^{j+1}_{\text{ada}} = \text{CTF}_j\left(\hat{F}^j_{\text{ada}}, F^j_{\text{spa}}, F^j_{\text{spe}}\right) \tag{6}$$

where $\text{CTF}_j(\cdot,\cdot,\cdot)$ denotes the token fusioner and F^{j+1}_{ada} is the input feature for the next adapter.

Next, the token injector module CTI_j injects the spatial and spectral priors obtained from the adapter back into the spatial feature extraction branch of HyperSIGMA [18], further enhancing its feature representation capability. The output feature $\hat{F}^j_{\text{spa}}$ is expressed as

$$\hat{F}^j_{\text{spa}} = \text{CTI}_j\left(F^{j+1}_{\text{ada}}, F^j_{\text{spe}}, F^j_{\text{spa}}\right). \tag{7}$$

Finally, the reshaped output feature from the last adapter is fed into the upsampling module for spatial resolution enhancement. To avoid an excessive number of parameters during fine-tuning, tokens are input into the adapter only once every t layers.

1) Spatial–Spectral Coupled Attention Module: The SSCAM is designed to simultaneously model the long-range dependencies in both the spatial and spectral dimensions of HSIs, thereby improving the SR reconstruction performance. This module includes both spatial attention and spectral attention mechanisms. The spatial attention mechanism enhances the spatial information of the image by calculating the correlation between pixels in the spatial dimension, thereby improving spatial resolution; the spectral attention mechanism, on the other hand, focuses on the relationships between different spectral bands, ensuring that the spectral features of the image are preserved. Initially, the long-range dependencies in both the spatial and spectral dimensions are modeled separately using self-attention mechanisms.

The formulas for the spatial feature F_{spa} and spectral feature F_{spe} are given by

$$F_{\text{spa}} = \text{Attention}\left(Q_{\text{spa}}, K_{\text{spa}}, V_{\text{spa}}\right) \tag{8}$$

$$F_{\text{spe}} = \text{Attention}\left(Q_{\text{spe}}, K_{\text{spe}}, V_{\text{spe}}\right) \tag{9}$$

where Q_{spa}, K_{spa}, and V_{spa} are the query, key, and value for the spatial features, respectively; and Q_{spe}, K_{spe}, and V_{spe} are the query, key, and value for the spectral features, respectively.

Next, the spatial and spectral features are further optimized through the cross-attention mechanism. In this mechanism, spatial information and spectral information are treated as both queries and keys. Specifically, spatial features are used as both query and key, while spectral features serve as both query and key. The cross-attention calculation is given by

$$A_{\text{cross}} = \text{Softmax}\left(\frac{Q_{\text{cross}}K^T_{\text{cross}}}{\sqrt{d_k}}\right) \tag{10}$$

where Q_{cross} and K_{cross} are the query and key for the spatial and spectral features, respectively, and d_k is the dimension of the key. The cross-attention is used to weight the spatial and spectral features, generating an enhanced output.

Finally, the SSCAM weights and fuses the spatial and spectral features to generate the final spatial–spectral feature output

$$F_{\text{fusion}} = \lambda_{\text{spa}}F_{\text{spa}} + \lambda_{\text{spe}}F_{\text{spe}} \tag{11}$$

where λ_{spa} and λ_{spe} are the coefficients that adjust the weight of spatial and spectral features, respectively. In this way, SSCAM can simultaneously optimize spatial details and spectral information during SR reconstruction, effectively improving the image quality while avoiding spectral distortion during spatial resolution enhancement.

2) Token Fusioner: The token fusioner (CTF) aims to effectively fuse features from both the spatial and spectral branches, optimizing the model's multimodal feature representation capability, as shown in Fig. 3. The core idea of the CTF module is to weight and fuse the spatial and spectral features, thus enhancing the feature representation ability of each adapter when processing HSIs. In the CTF module, spatial features F^j_{spa} and spectral features F^j_{spe} are element-wise multiplied through a weighted network $w_j(\cdot)$, generating the fused features. The weighted network allows the fusion of spatial and spectral features to be dynamically adjusted, adapting to differences in input features, resulting in more precise feature representation. This process is expressed by the following equation:

$$\tilde{F}^j_{\text{spa}} = F^j_{\text{spa}} \cdot w_j\left(F^j_{\text{spe}}\right) + F^j_{\text{spa}} \tag{12}$$

where $\tilde{F}^j_{\text{spa}}$ represents the weighted spatial features and $w_j(\cdot)$ is the weighted network that controls the fusion strength between spatial and spectral features.

The fused spatial features $\tilde{F}^j_{\text{spa}}$ are then further optimized through cross-attention by interacting with the adapter features

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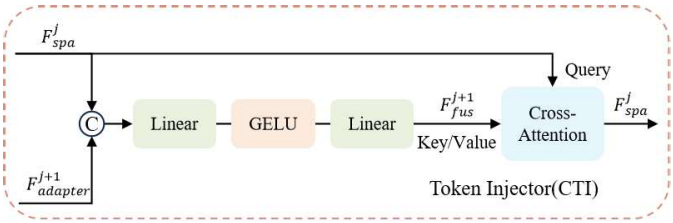


Fig. 4. Architectures of the token injector module.

F^j_{ada}, enhancing the final feature representation. In the cross-attention mechanism, F^j_{ada} serves as the query (query), while $\tilde{F}^j_{\text{spa}}$ serves as both the key (key) and value (value). The cross-attention calculation is expressed as

$$\hat{F}^j_{\text{ada}} = \text{Attention}\left(F^j_{\text{ada}}, \tilde{F}^j_{\text{spa}}\right) + F^j_{\text{ada}}. \tag{13}$$

This process utilizes a multihead cross-attention mechanism to generate the enhanced output feature $\hat{F}^j_{\text{ada}}$ by weighting the fused spatial and adapter features.

Subsequently, $\hat{F}^j_{\text{ada}}$ is passed through a feedforward network (FFN) to produce the input feature for the next adapter. This process is expressed as follows:

$$F^{j+1}_{\text{ada}} = \hat{F}^j_{\text{ada}} + \text{FFN}\left(\hat{F}^j_{\text{ada}}\right). \tag{14}$$

Finally, the output F^{j+1}_{ada} serves as the input feature for the next adapter, ensuring that the features are passed through the network for further processing.

The advantage of the CTF module lies in its flexible feature-fusion strategy, which dynamically adjusts the fusion of spatial and spectral features based on the characteristics of the input features, thus improving SR performance. Through this weighted fusion and cross-attention mechanism, the CTF module effectively captures the complex relationships between spatial and spectral features, ultimately improving image quality.

3) Token Injector: The token injector (CTI) aims to inject spatial and spectral prior information from the adapter back into the spatial feature extraction branch of HyperSIGMA [18], further enhancing the model's feature representation ability, as shown in Fig. 4. By interacting the adapter output with the spatial features, the CTI module optimizes the model's capability to capture spatial features, thus improving the SR reconstruction performance. In the CTI module, the output feature F^j_{ada} from the adapter and the spatial feature F^j_{spa} are first concatenated along the feature dimension and projected into a specific dimension k using a linear layer, resulting in F^j_{fus} as the query for the cross-attention. This process is expressed as

$$F^j_{\text{fus}} = \text{Linear}\left(\text{Concat}\left(F^j_{\text{ada}}, F^j_{\text{spa}}\right)\right). \tag{15}$$

Building on this, the cross-attention mechanism further optimizes the interaction between spatial features and adapter features. Specifically, the spatial features F^j_{spa} from the jth layer of HyperSIGMA [18] are used as both the key and value for the cross-attention, and the cross-attention calculation is expressed as

$$F^j_{\text{spa}} = s_j \cdot \text{Attention}\left(F^j_{\text{spa}}, F^j_{\text{fus}}\right) + F^j_{\text{spa}} \tag{16}$$

where s_j is a learnable parameter initialized to 0. Through the cross-attention mechanism, the resulting F^{spa}_j is injected back into the spatial feature extraction branch of HyperSIGMA [18], enhancing the model's feature extraction capability for the SR task.

The advantage of the CTI module lies in its ability to effectively inject spatial and spectral prior information from the adapter back into HyperSIGMA [18], enabling the model to better capture spatial features and improve HSI SR reconstruction. Through this process, the CTI module not only enhances the model's ability to model spatial features but also improves the model's adaptability to complex HSIs.

D. Spatial–Spectral Loss Function

In HSI SR tasks, the L1 loss function is commonly used and has been shown to achieve good results in various studies [40]. The L1 loss minimizes the pixel-wise differences between the reconstructed and ground truth images, effectively recovering spatial details. Compared to the L2 loss, the L1 loss exhibits greater balance in the error distribution and typically results in faster convergence. However, while L2 loss can effectively reduce pixel-wise errors, it often leads to excessive smoothing, which can suppress fine image details. Although L1 loss performs well for SR of natural images, it neglects the spectral characteristics of HSIs, which may lead to spectral distortion. Therefore, to address the limitations of L1 loss in HSI SR, this study introduces spectral angle mapping (SAM) loss to ensure spectral consistency while preserving spatial details.

Specifically, the L1 loss function is used to measure the pixel-wise differences between the reconstructed HSI and the original HR HSI. It is computed as

$$L_1(\theta) = \frac{1}{N}\sum_{n=1}^{N}\left|I^n_{\text{hr}} - I^n_{\text{sr}}\right|_1 \tag{17}$$

where N is the number of images in the training batch and I^n_{hr} and I^n_{sr} represent the nth original HR HSI and its reconstructed counterpart, respectively. Although L1 loss performs well in super-resolving natural images, it does not account for the spectral characteristics of HSIs, which can lead to spectral distortion. To maintain spectral consistency, this study introduces SAM loss, which is computed as

$$L_{\text{sam}}(\theta) = \frac{1}{N}\sum_{n=1}^{N}\arccos\left(\frac{I^n_{\text{sr}} \cdot I^n_{\text{hr}}}{\left\|I^n_{\text{sr}}\right\|_2 \left\|I^n_{\text{hr}}\right\|_2}\right) \tag{18}$$

where $\cdot$ denotes the dot product and $\|\cdot\|_2$ represents the L2 norm of a vector. SAM loss measures the spectral difference between the reconstructed and original images by computing the angle between their spectral vectors, thus ensuring spectral consistency.

In addition, to further improve the clarity of the reconstructed images, this study introduces gradient loss, inspired by prior research [22]. Gradient loss enhances image sharpness by calculating the differences between adjacent pixels. The gradient of the HSI I is represented as

$$\nabla I = (\nabla_h I, \nabla_w I, \nabla_l I) \tag{19}$$

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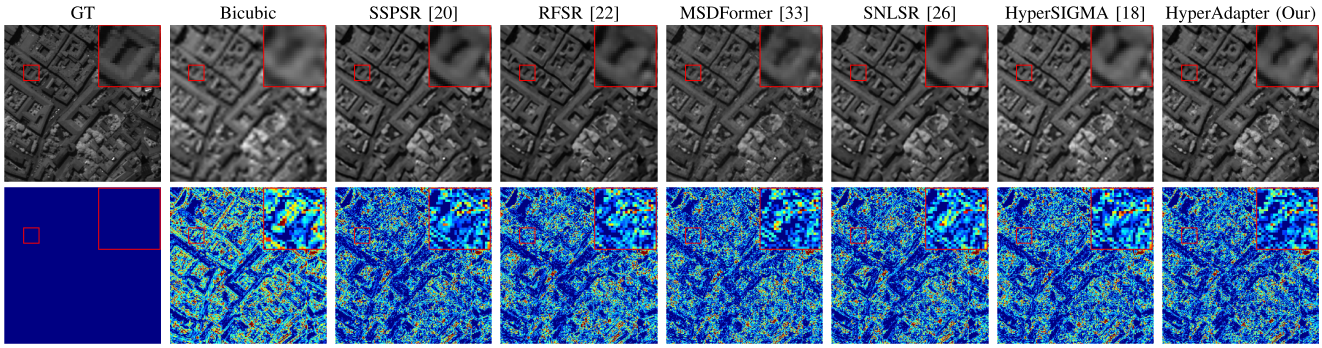


Fig. 5. Visual comparison of different methods for 4× SR on the Pavia dataset. (Top row) Reconstruction results of the 64th spectral band and (bottom row) error maps.

where $M(I)$ is the operator used to obtain the gradient map and ∇_h, ∇_w, and ∇_l represent the gradient computation functions along the horizontal, vertical, and spectral dimensions, respectively. Gradient loss minimizes the distance between the gradient maps of the reconstructed and original images, thus improving the local clarity of the image. The gradient loss is computed as

$$L_{\text{gra}}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \left\| M\left(I_{\text{sr}}^{n}\right) - M\left(I_{\text{hr}}^{n}\right) \right\|_1 \tag{20}$$

where $\|\cdot\|_1$ denotes the L1 norm. This loss reduces the distance between the gradient maps of the reconstructed and ground truth images, enhancing the local sharpness of the image.

Thus, the overall loss function used in this study is the weighted combination of L1 loss, SAM loss, and gradient loss, which is expressed as follows:

$$L(\theta) = L_1(\theta) + \alpha \cdot L_{\text{sam}}(\theta) + \beta \cdot L_{\text{gra}}(\theta) \tag{21}$$

where $\alpha = 0.5$ and $\beta = 0.1$ are the weights that balance the SAM loss and gradient loss. This weighted loss function allows the model to optimize spatial details, spectral consistency, and local clarity simultaneously, thereby achieving superior performance in HSI SR.

IV. Experiments

A. Experimental Setup

To validate the effectiveness of the proposed method, we selected three benchmark datasets commonly used in hyperspectral remote sensing for the experiment: the Pavia Center dataset [51], the Chikusei dataset [52], and the Houston dataset.

The Pavia Center dataset comprises 102 spectral bands with a spatial resolution of 1.3 m, and the original image dimensions are 1096 × 1096 pixels. Due to the presence of invalid regions in the data, this study performed precise cropping of the original image following the SSPSR method [20], resulting in a valid region size of 1096 × 715 pixels. For dataset partitioning, four nonoverlapping 256 × 256 pixel subimages were extracted from the left side of the image as the test set, while the remaining area was used as the training set.

The Chikusei dataset comprises 128 spectral bands with original spatial dimensions of 2517 × 2335 pixels. To address invalid regions in the data, this study employed the SSPSR method [20] for preprocessing, wherein precise cropping was applied to retain the central valid area, resulting in a refined image size of 2304 × 2048 pixels. During the dataset partitioning phase, four nonoverlapping 512 × 512 pixel subimages were extracted from the top portion of the image to form the test set, while the remaining area was allocated for training purposes.

The Houston dataset comprises 48 spectral bands with original spatial dimensions of 4172 × 1202 pixels. In the experimental design of this study, we adopted a systematic partitioning strategy: four nonoverlapping 256 × 256 pixel subimages were extracted from the top portion of the image to constitute the test set, while the remaining area was utilized as the training set.

In the experiments, all parameters of both the spatial feature branch and the spectral feature branch of the base model Hypersigma were kept frozen, with only the newly added adapter network along with the feature-fusion module and upsampling module in Hypersigma being trained end-to-end. The adapter network was trained using datasets such as the Pavia Center Dataset, Chikusei Dataset, and Houston Dataset, with the training and test sets partitioned according to classical configurations. The original HR image is treated as the ground truth reference, and the corresponding LR image is generated using bicubic downsampling. Data augmentation strategies, including random horizontal/vertical flipping, were employed during the training process. The training process uses the Adam optimizer with a total of 1000 epochs. The initial learning rate is set to 6×10^{-5}, and the batch size is set to 1. The learning rate is decayed by a factor of 0.5 every 150 epochs. The proposed method is implemented using the PyTorch framework on a single NVIDIA RTX 3090 GPU.

To evaluate the experimental results, we used several common image quality metrics, including PSNR, SSIM, SAM, CC, and ERAGS, to comprehensively assess the compression performance. Additionally, we visually demonstrated the image reconstruction effects and errors of different methods to provide a more intuitive comparison of their performance.

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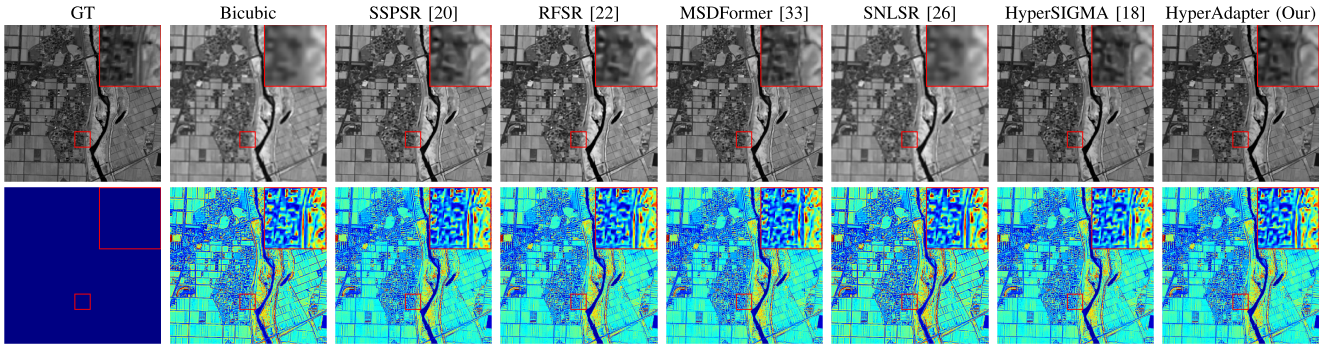


Fig. 6. Visual comparison of different methods for 2× SR on the Chikusei dataset. (Top row) Reconstruction results of the 100th spectral band and (bottom row) error maps.

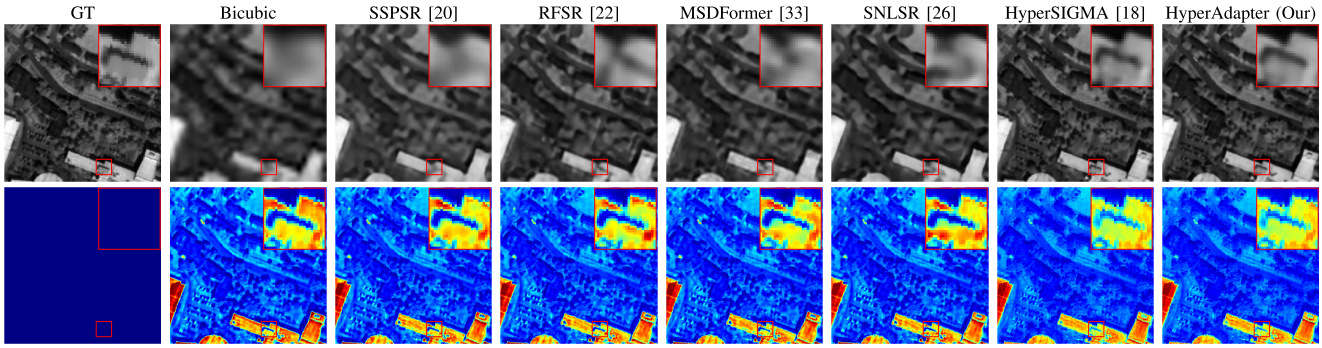


Fig. 7. Visual comparison of different methods for ×8 SR on the Houston dataset. (Top row) Reconstruction results of the 30th spectral band and (bottom row) error maps.

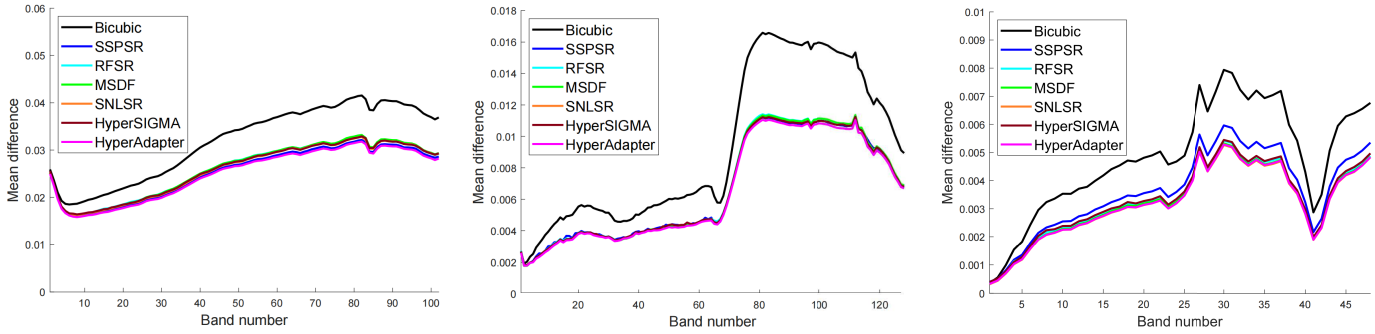


Fig. 8. Comparative spectral reconstruction across datasets. (a) Pavia, (b) Chikusei, and (c) Houston, demonstrating the average spectral band errors between reconstructed results and ground truth for various SR methods.

B. Comparison With the SOTA Methods

In order to systematically evaluate the performance of the proposed method, HyperAdapter was compared with six existing HSI SR methods, including Bicubic, SSPSR [20], RFSR [22], MSDFormer [33], SNLSR [26], and HyperSIGMA [18].

Experimental results on the Pavia Center dataset, as summarized in Table I, demonstrate that the performance of HyperAdapter has achieved SOTA results. In the 2× SR task, it surpasses the baseline model HyperSIGMA [18] by 0.3 dB in PSNR, exceeding all other compared methods by a considerable margin while also demonstrating superior performance in SSIM and ERGAS. Its efficacy is further exemplified in the more challenging 4× SR task, where it exhibits a PSNR gain of 0.25 dB and an SSIM improvement of 0.011 over HyperSIGMA [18]. Moreover, notable enhancements are observed across additional metrics, including SAM, CC, and ERGAS. Fig. 5 visually compares the reconstruction outcomes of different methods and their corresponding residual maps relative to the ground truth. These visualizations confirm that the proposed HyperAdapter method effectively restores fine spatial details with higher fidelity. Furthermore, as shown in Fig. 8(a), while HyperSIGMA's spectral sampling strategy exhibits certain limitations in spectral reconstruction on this dataset, HyperAdapter achieves lower spectral reconstruction errors across most bands, underscoring its robustness in maintaining spectral consistency. This superior performance stems

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TABLE I

QUANTITATIVE EVALUATION WITH SOTA METHODS FOR HSI SR AT MULTIPLE SCALES (2×, 4×, AND 8×) ON PAVIA CENTER DATASETS (PSNR↑, SSIM↑, SAM↓, CC↑, AND ERGAS↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	×2					×4					×8				
	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓
Bicubic	32.19	0.912	4.56	0.955	8.51	27.66	0.725	6.45	0.872	7.05	24.77	0.481	8.16	0.734	4.90
SSPSR [20]	34.89	0.951	4.10	0.974	6.36	29.06	0.811	5.81	0.907	6.02	25.19	0.530	8.16	0.758	4.67
RFSR [22]	35.20	0.954	3.82	0.976	6.15	28.94	0.806	5.86	0.904	6.11	25.21	0.527	7.90	0.759	4.70
MSDFormer [33]	35.56	0.959	**3.65**	0.977	5.92	28.89	0.809	5.77	0.903	6.14	25.20	**0.534**	7.86	0.758	4.67
SNLSR [26]	35.51	0.957	3.75	0.977	5.96	28.90	0.805	5.75	0.903	6.14	25.14	0.527	8.14	0.755	4.69
HyperSIGMA [18]	35.51	0.957	3.81	0.977	5.93	28.95	0.806	5.84	0.904	6.11	25.20	0.533	**7.82**	**0.761**	4.68
HyperAdapter	**35.81**	**0.960**	**3.65**	**0.978**	**5.76**	**29.20**	**0.817**	**5.64**	**0.910**	**5.93**	**25.22**	**0.534**	7.85	**0.761**	**4.66**

TABLE II

QUANTITATIVE EVALUATION WITH SOTA METHODS FOR HSI SR AT MULTIPLE SCALES (2×, 4×, AND 8×) ON CHIKUSEI DATASETS (PSNR↑, SSIM↑, SAM↓, CC↑, AND ERGAS↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	×2					×4					×8				
	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓
Bicubic	43.21	0.972	1.79	0.978	7.20	37.64	0.895	3.40	0.921	6.76	34.51	0.807	5.04	0.831	4.85
SSPSR [20]	47.34	0.989	1.22	0.990	4.67	39.99	0.940	2.51	**0.953**	5.18	35.38	0.838	4.44	0.864	4.39
RFSR [22]	47.55	0.990	1.17	**0.991**	4.55	**40.02**	0.940	2.45	**0.953**	**5.16**	35.41	0.842	4.28	0.864	4.39
MSDFormer [33]	47.57	**0.991**	1.16	0.990	4.51	39.92	0.939	2.43	0.952	5.18	35.57	0.845	4.21	0.869	4.21
SNLSR [26]	43.21	0.972	1.79	0.978	7.20	37.64	0.895	3.40	0.921	7.86	35.60	0.847	4.08	0.869	4.25
HyperSIGMA [18]	47.37	0.989	1.22	**0.991**	4.66	39.98	0.940	2.44	**0.953**	5.18	35.69	0.850	4.07	0.872	4.23
HyperAdapter	**47.74**	0.990	**1.13**	**0.991**	**4.46**	**40.02**	**0.941**	**2.40**	**0.953**	**5.16**	**35.75**	**0.851**	**3.99**	**0.874**	**4.20**

TABLE III

QUANTITATIVE EVALUATION WITH SOTA METHODS FOR HSI SR AT MULTIPLE SCALES (2×, 4×, AND 8×) ON HOUSTON DATASETS (PSNR↑, SSIM↑, SAM↓, CC↑, AND ERGAS↓). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	×2					×4					×8				
	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓
Bicubic	49.67	0.992	1.21	0.995	2.60	43.08	0.963	2.45	0.963	2.79	38.11	0.923	4.53	0.925	4.95
SSPSR [20]	52.23	**0.996**	0.98	**0.997**	1.91	45.62	0.979	1.91	0.986	2.06	39.57	0.926	3.85	0.945	2.05
RFSR [22]	52.75	**0.996**	0.90	**0.997**	1.79	46.39	0.982	1.67	0.988	1.88	39.75	0.925	3.58	0.947	2.04
MSDFormer [33]	52.81	**0.996**	0.93	**0.997**	1.79	46.51	**0.983**	**1.63**	**0.989**	1.86	39.96	0.929	3.48	0.949	1.98
SNLSR [26]	52.86	**0.996**	0.90	**0.997**	1.78	46.48	**0.983**	1.66	**0.989**	1.87	40.05	0.930	3.46	0.949	1.97
HyperSIGMA [18]	52.40	**0.996**	0.99	**0.997**	1.84	46.29	0.982	1.74	0.988	1.89	40.12	0.930	3.49	0.950	1.95
HyperAdapter	**52.98**	**0.996**	**0.89**	**0.997**	**1.75**	**46.72**	**0.983**	1.65	**0.989**	**1.81**	**40.22**	**0.932**	**3.35**	**0.952**	**1.92**

from the novel architecture of HyperAdapter, which facilitates more effective integration of spatial and spectral features throughout the reconstruction process.

Experimental results on the Chikusei dataset, as summarized in Table II, demonstrate that the performance of HyperAdapter has achieved SOTA results. It is noteworthy that while the inherent spectral sampling strategy of the base model imposes certain limitations on the SAM metric for the Chikusei dataset, HyperAdapter nonetheless achieves superior overall performance compared to all alternative methods. For detailed visual analysis, Fig. 6 provides residual maps contrasting the reconstructed images with the ground truth, alongside visual outcomes from different methods within selected spectral bands. Furthermore, Fig. 8(b) quantitatively compares the average reconstruction errors across the entire spectral range for each method. Collectively, both the visual reconstruction results and the spectral consistency analysis validate that HyperAdapter proficiently recovers fine image details while maintaining high spectral fidelity.

Experimental results on the Houston dataset, as summarized in Table III, demonstrate that the performance of HyperAdapter has achieved SOTA results. At lower SR scales (2× and 4×), HyperAdapter achieves SSIM and CC metrics comparable to those of previous methods, while demonstrating a significant performance advantage in PSNR, with improvements of more than 0.58 and 0.43 dB over the baseline model at 2× and 4× SR. Furthermore, in the ×8 SR task, HyperAdapter consistently outperforms all other methods across every evaluation metric. For qualitative assessment, Fig. 7 displays residual maps between the reconstructed and ground truth images, alongside visual comparisons across selected spectral bands. Quantitative analysis in Fig. 8(c) illustrates the per-method average error across all spectral bands. As clearly evidenced, the proposed HyperAdapter yields the lowest error spectrum-wide, outperforming all baseline methods and confirming its superior accuracy in spectral reconstruction and enhanced effectiveness in preserving spectral consistency for hyperspectral imagery.

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TABLE IV
MODULE-WISE ABLATION STUDY OF 4× SR PERFORMANCE ON THE HOUSTON DATASET

Method	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓
w/o SSPM	46.31	0.982	1.69	0.988	1.90
w/o CTF	46.61	0.982	1.67	**0.989**	1.83
w/o CTI	46.47	0.982	1.67	**0.989**	1.87
ours	**46.72**	**0.983**	**1.65**	**0.989**	**1.81**

TABLE V
IMPACT OF THE NUMBER OF ADAPTER MODULES (N) ON THE SR PERFORMANCE AT 4× SCALING ON THE HOUSTON DATASET

N	PSNR↑	SSIM↑	SAM↓	CC↑	ERGAS↓
1	45.56	0.979	1.85	0.986	2.08
2	46.32	0.982	1.70	0.988	1.90
3	46.52	**0.983**	1.68	**0.989**	1.86
4	**46.72**	**0.983**	**1.65**	**0.989**	**1.81**
6	46.69	0.982	1.70	0.988	1.90
12	5.82	0.003	49.73	0.06	317.74

C. Ablation Study

To further validate the effectiveness of each module in HyperAdapter, we conducted the corresponding ablation experiments on the Houston dataset to analyze the contribution of each module to the model's performance.

1) Spatial–Spectral Prior Module: The proposed SSPM aims to effectively model the local spatial context of HSIs, thereby enhancing the Transformer model's ability to capture local information. The experimental results are shown in Table IV, where "HyperAdapter" represents our proposed method and "w/o SSPM" indicates replacing the SSPM with a commonly used convolution module. In this ablation experiment, all other modules remain unchanged except for the ablated module. The results show that after replacing the SSPM, spatial metrics (PSNR and ERGAS) significantly decrease, and the spectral metric (SAM) also shows a certain degree of decline. This indicates that the SSPM indeed enhances the spatial local feature extraction capability of HyperSIGMA [18] and maintains spectral consistency to some extent.

2) Number of Adapters: The adapter module proposed in this method effectively enhances the features of HyperSIGMA [18], improving its performance in HSI SR tasks. This experiment investigates the effect of the number of adapters (N) on the SR reconstruction performance. The number of adapters plays a crucial role in this method and directly affects the model's SR performance. As observed in Table V, with an increasing number of adapters, the model's SR performance gradually improves. Specifically, when N increases from 1 to 4, metrics such as PSNR and SSIM show significant improvements, indicating that the introduction of adapters effectively enhances the model's ability to extract HSI features. However, when the number of adapters exceeds a certain threshold (e.g., $N = 6$ or higher), the model's performance starts to decline. In particular, when N reaches 12, the performance of the model deteriorates drastically, and all metrics almost fail. This phenomenon may be due to the excessive number of adapters interfering with the feature extraction process of HyperSIGMA [18], preventing the model from fully exploiting its powerful feature extraction capabilities, thus affecting the SR performance.

3) Token Fusion Module: The design of the CTF module aims to effectively fuse the spatial and spectral features from HyperSIGMA [18] to enhance the model's feature expression capabilities. To verify the effectiveness of this module, an ablation experiment was conducted by replacing the CTF module with a commonly used residual module while keeping all other modules unchanged. The experimental results are shown in Table IV, where "HyperAdapter" represents our proposed method and "w/o CTF" indicates the model without the CTF module. The results show that after removing the CTF module, the model's SR performance decreases across multiple metrics. These results indicate that the CTF module plays a role in the feature-fusion process and effectively integrates spatial and spectral features, although with a relatively small impact, which still influences the model's feature representation capability.

4) Token Injection Module: The design of the CTI module aims to inject spatial and spectral prior information from the adapter into the spatial feature extraction branch of HyperSIGMA [18] to further enhance the ability to represent features. To verify the effectiveness of this module, an ablation experiment was conducted by replacing the CTI module with a commonly used residual module while keeping all other modules unchanged. The experimental results are also shown in Table IV, where "w/o CTI" represents the model without the CTI module. After removing the CTI module, the model's SR performance decreases across multiple.

V. CONCLUSION

In this article, we proposed a novel HSI SR method called HyperAdapter, which is based on adapter-enhanced foundational model, HyperSIGMA [18]. To address the architectural limitations of foundation models, the proposed method introduces both SSPMs and adapter modules. This approach effectively leverages the rich prior knowledge embedded in pretrained models while significantly enhancing the model's capability to fuse spatial and spectral features, thereby improving SR reconstruction performance. Extensive experimental results on publicly available HSI datasets, including Houston, Pavia Center, and Chikusei, demonstrate that HyperAdapter consistently outperforms state-of-the-art methods, particularly in high-magnification tasks. The method excels in recovering fine spatial details while maintaining spectral consistency, showcasing its robustness and practical applicability. Furthermore, ablation studies validate the effectiveness of each component, especially the spatial–spectral prior and adapter modules, in enhancing the overall performance of the model. Future research directions should focus on in-depth investigation of spatial–spectral feature-fusion mechanisms and architectural optimization strategies for foundation models, which may lead to substantial advancements in HSI SR tasks. The proposed HyperAdapter framework presents an innovative

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approach to addressing current limitations in HSI reconstruction, making significant contributions to the advancement of this research field.

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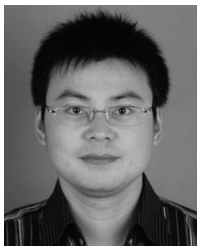
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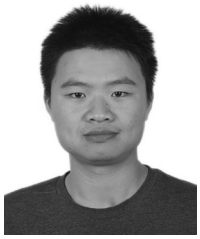
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